

CCNA Mini Project — Server & Service

DNS-Based Web Hosting Simulation: Hotstar & Netflix over a Routed Network

Lab Environment: GNS3 / EVE-NG with QEMU virtual machines

Project Repository: github.com/Kaviyarasu-K07/CCNA

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Objective: This project simulates a small routed network in which two internal web servers, styled after Hotstar and Netflix, are hosted in a datacenter segment and made reachable from a client PC first by IP address and then by domain name, using a dedicated DNS server. The project demonstrates static IP addressing, inter-VLAN style routing between three subnets on a single router, installation and verification of IIS on Windows Server 2016, hosting a static page on a Linux web server, and configuring a Windows DNS server with custom forward-lookup host records.

1. Project Overview & Network Topology

The lab consists of a single router connecting three subnets: a client subnet (50.0.0.0/8) holding the test PC, a datacenter subnet (172.16.0.0/16) holding the two web servers, and a DNS subnet (8.0.0.0/8) holding the DNS server. The router performs the routing between all three subnets over its three Ethernet interfaces. The Linux server represents "Hotstar" and the Windows Server 2016 (IIS) machine represents "Netflix".

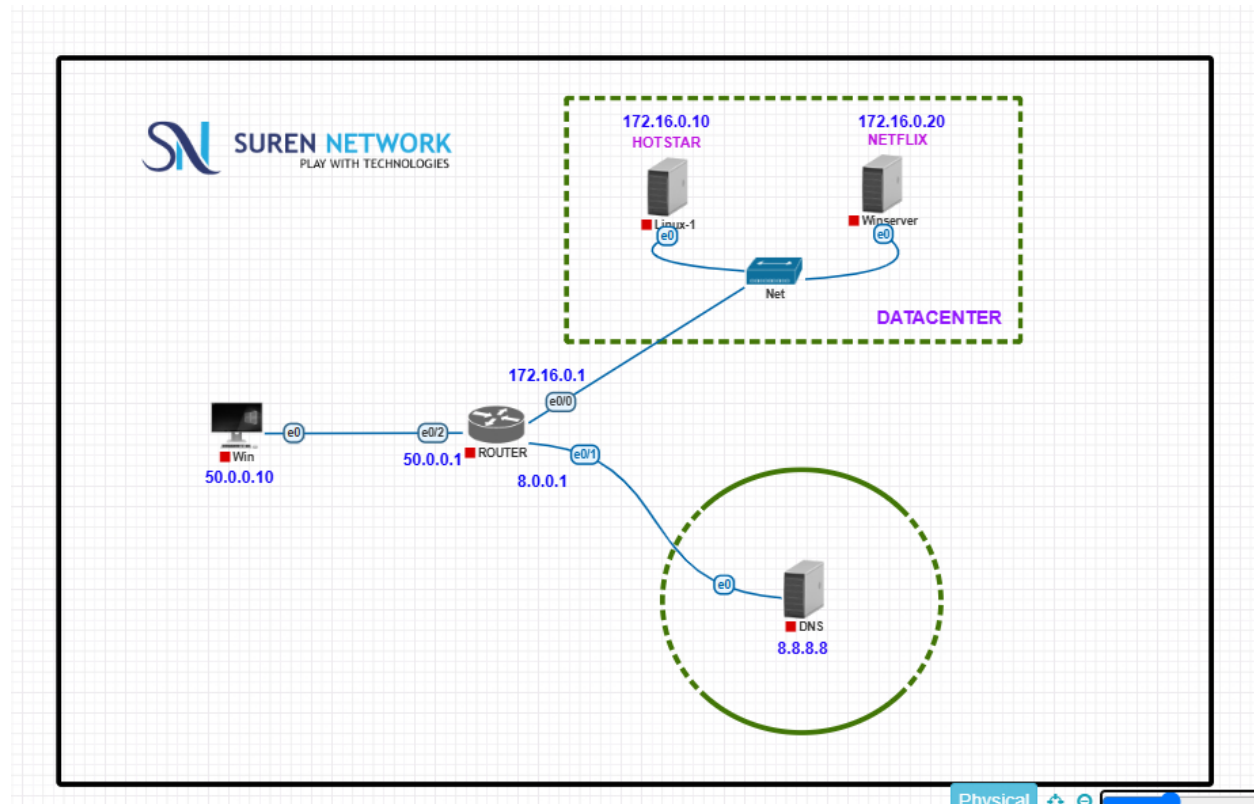


Figure 1 — Logical network topology (Suren Network lab)

2. IP Addressing Plan

The table below summarizes every interface configured in this lab, matching the addressing scheme visible across the router and end-device configuration screens.

Device	Role	Interface	IP Address	Subnet Mask	Default Gateway
ROUTER	Gateway — Datacenter	e0/0	172.16.0.1	255.255.0.0	—
ROUTER	Gateway — DNS subnet	e0/1	8.0.0.1	255.0.0.0	—
ROUTER	Gateway — Client subnet	e0/2	50.0.0.1	255.0.0.0	—
Linux-1 (HOTSTAR)	Web server	eth0	172.16.0.10	255.255.0.0	172.16.0.1
Winserver (NETFLIX)	Web server (IIS)	Ethernet	172.16.0.20	255.255.0.0	172.16.0.1
DNS	DNS server	Ethernet	8.8.8.8	255.0.0.0	8.0.0.1
Win	Client PC	Ethernet	50.0.0.10	255.0.0.0	50.0.0.1

The client's preferred DNS server is later set to 8.8.8.8 — the internal lab DNS server's address, not the public Google DNS service — so that the client resolves the two custom domain names through this lab's own DNS zones.

3. Router Configuration

3.1 Configuring the Router Interfaces

Interfaces e0/0 and e0/1 are addressed from global configuration mode with `ip address` and brought up with `no shutdown`. The `show ip interface` brief output confirms all three Ethernet interfaces (e0/0, e0/1, e0/2) are up/up, and `show ip route` confirms the router has installed a directly-connected route for each of the three subnets — this is what allows traffic to pass between the client, the datacenter, and the DNS subnet.

```
ROUTER
ROUTER(config-if)#no sh
ROUTER(config-if)#exit
ROUTER(config)#
ROUTER(config)#int e0/0
ROUTER(config-if)#ip add 173
% Incomplete command.

ROUTER(config-if)#ip add 172.16.0.1 255.255.0.0
ROUTER(config-if)#no sh
ROUTER(config-if)#exit
ROUTER(config)#int e0/1
ROUTER(config-if)#ip add 8.0.0.1 255.0.0.0
ROUTER(config-if)#no sh
ROUTER(config-if)#exit
ROUTER(config)#exit
ROUTER#
*Aug 29 17:09:12.710: %SYS-5-CONFIG_I: Configured from console by console
ROUTER#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              172.16.0.1      YES manual up          up
Ethernet0/1              8.0.0.1         YES manual up          up
Ethernet0/2              50.0.0.1        YES manual up          up
Ethernet0/3              unassigned      YES TFTP  down        down
ROUTER#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      8.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       8.0.0.0/8 is directly connected, Ethernet0/1
L       8.0.0.1/32 is directly connected, Ethernet0/1
      50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       50.0.0.0/8 is directly connected, Ethernet0/2
L       50.0.0.1/32 is directly connected, Ethernet0/2
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, Ethernet0/0
L       172.16.0.1/32 is directly connected, Ethernet0/0
ROUTER#
```

Figure 3.1 — Configuring the Router Interfaces

4. Netflix Web Server Setup — Windows Server 2016 + IIS

4.1 Opening Server Manager

On the Windows Server 2016 VM that will act as the "Netflix" server, Server Manager is opened and **Add roles and features** is selected from the Quick Start panel to begin installing the web server role.

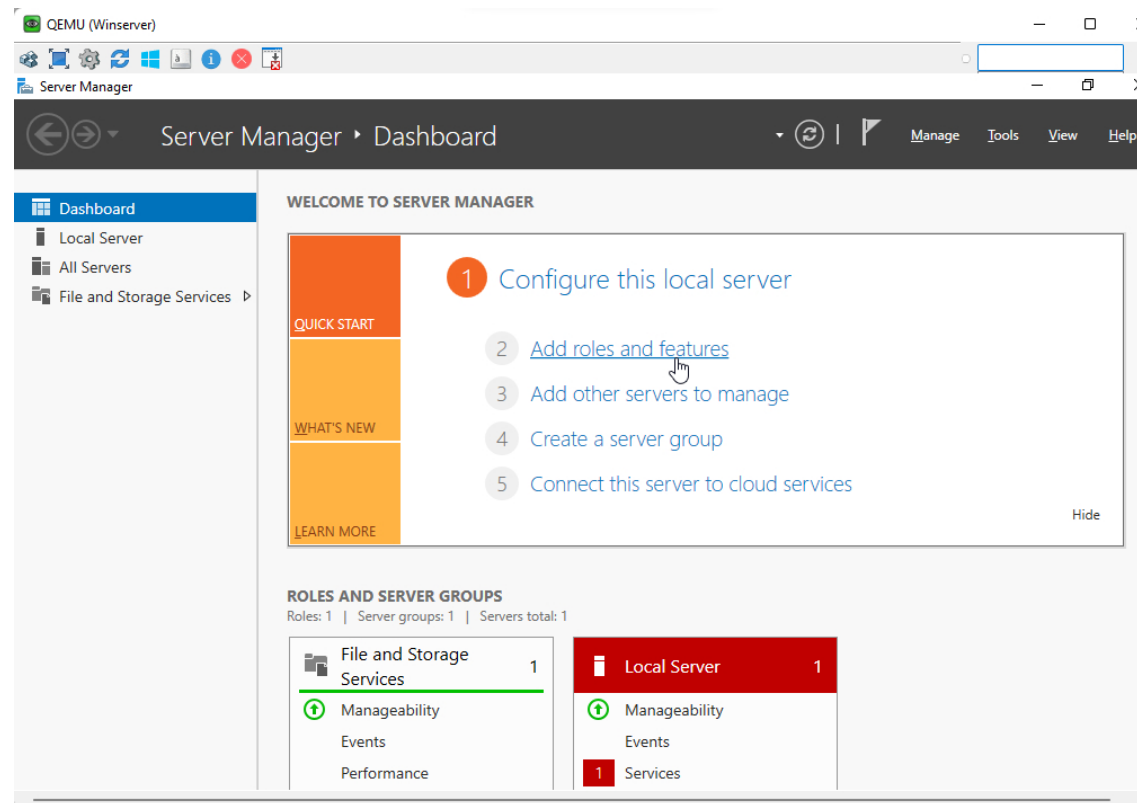


Figure 4.1 — Opening Server Manager

4.2 Selecting the Web Server (IIS) Role

In the Add Roles and Features Wizard, the **Web Server (IIS)** role is located in the roles list. Selecting it will install Internet Information Services, which is what will serve the Netflix-styled web page.

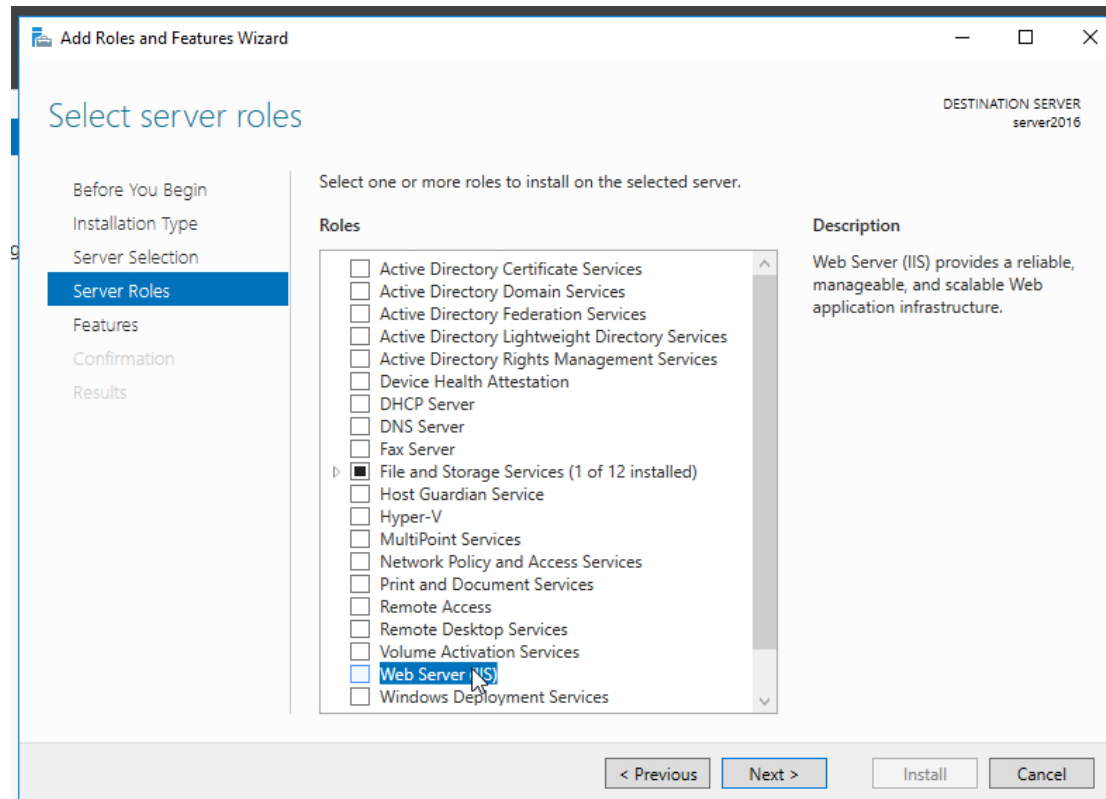


Figure 4.2 — Selecting the Web Server (IIS) Role

4.3 Confirming the Role Selection

The Web Server (IIS) checkbox is ticked, adding it to the pending installation. The wizard automatically includes the dependent management and web-server features.

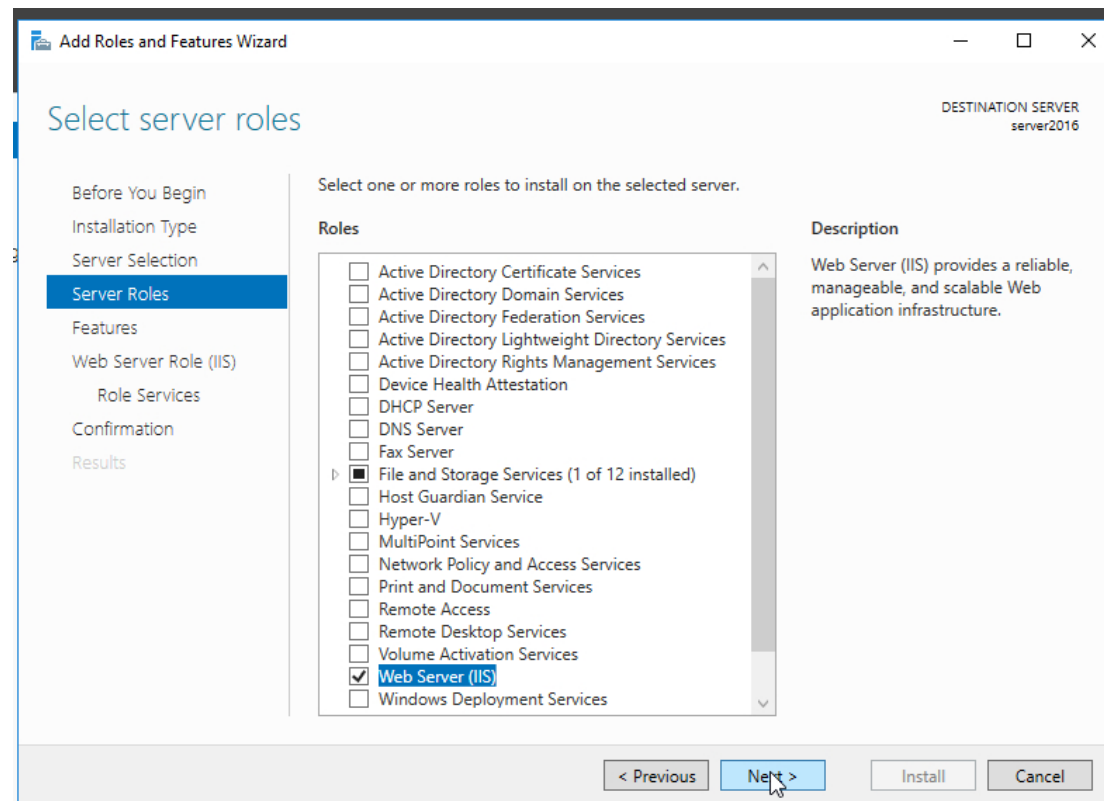


Figure 4.3 — Confirming the Role Selection

4.4 Reviewing the Installation Selections

The confirmation screen lists everything that will be installed: the IIS Management Console under Management Tools, and Common HTTP Features (Default Document, Directory Browsing, HTTP Errors, Static Content) under the Web Server role. Clicking **Install** begins the installation.

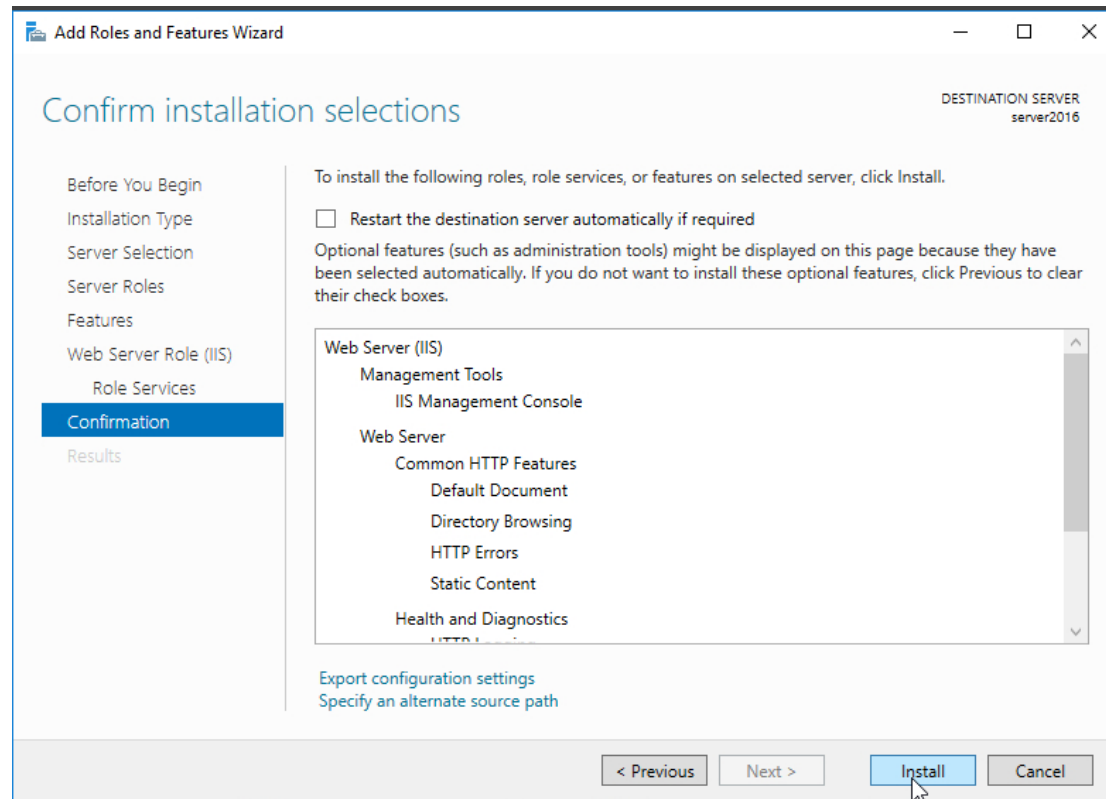


Figure 4.4 — Reviewing the Installation Selections

4.5 Installation Completed

The results page reports that the Web Server (IIS) feature installation succeeded on server2016. The server can now host and serve web content over HTTP.

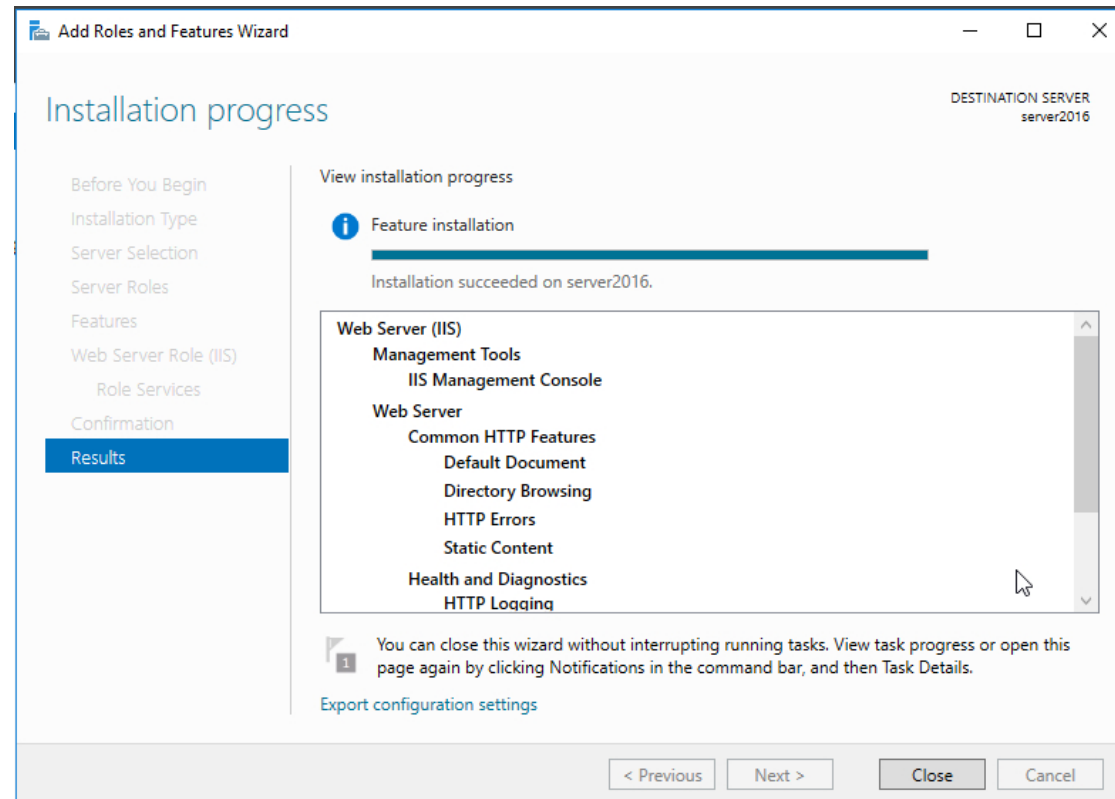


Figure 4.5 — Installation Completed

5. Hotstar Web Server Setup — Linux

5.1 Opening the Network Interface Configuration File

On the Linux-1 VM, the interface configuration file for eth0 is opened with `vi /etc/sysconfig/network-scripts/ifcfg-eth0` so that a static IP address can be assigned instead of relying on DHCP.

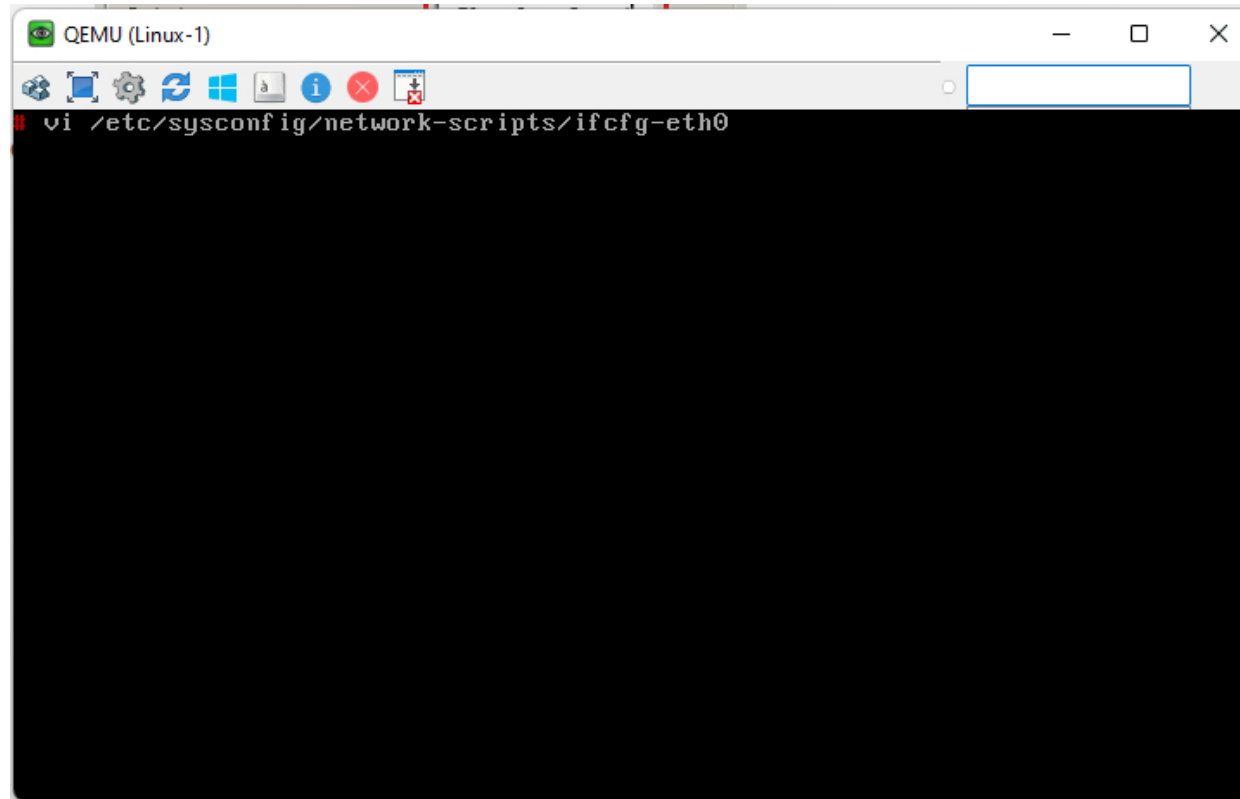


Figure 5.1 — Opening the Network Interface Configuration File

5.2 Assigning a Static IP to eth0

The file is edited to set IPADDRESS=172.16.0.10, CIDRLEN=16, NETWORK=172.16.0.0, NETMASK=255.255.0.0, GATEWAY=172.16.0.1, and DHCP=no, matching the addressing plan for the datacenter subnet.

A screenshot of a QEMU virtual machine window titled "QEMU (Linux-1)". The window has a standard Linux desktop environment with icons for file manager, settings, and other applications. A terminal window is open, displaying the following network configuration commands and their output:

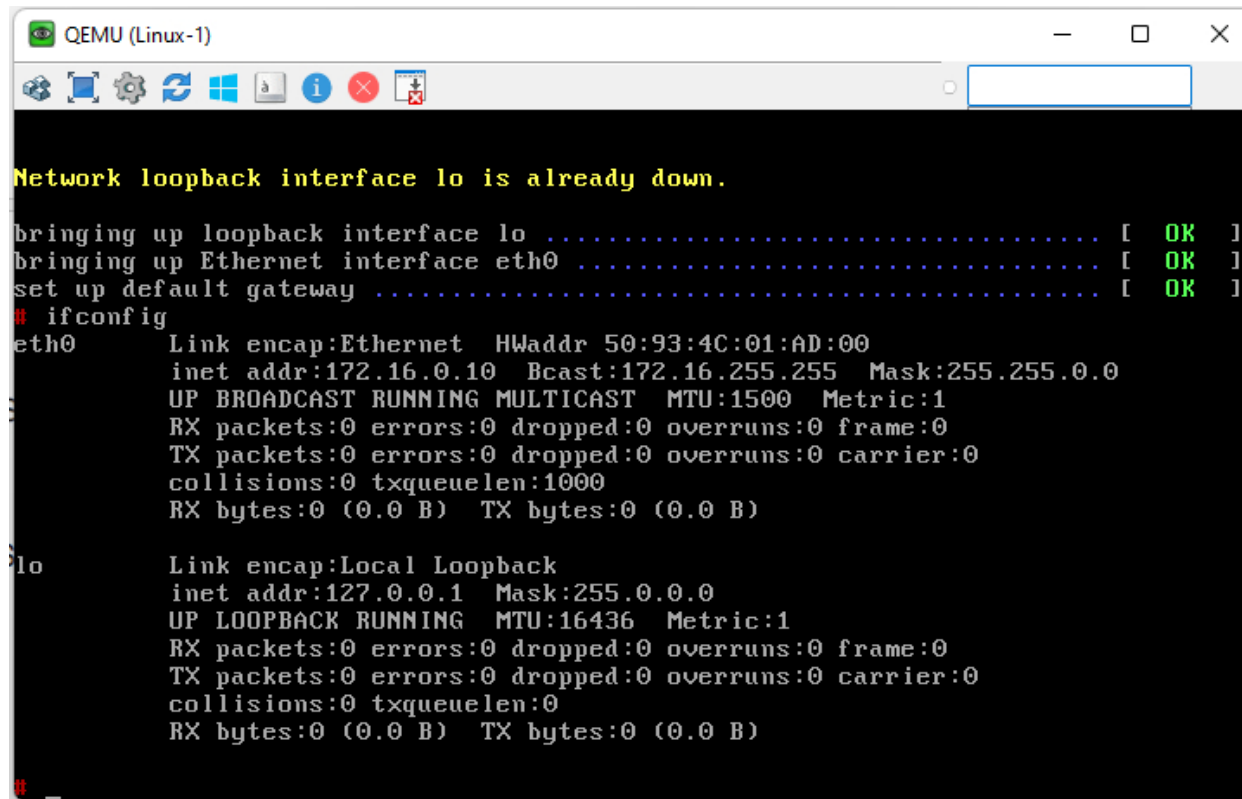
```
ENABLE=yes  
NAME=Ethernet  
IPADDRESS=172.16.0.10  
CIDRLEN=16  
NETWORK=172.16.0.0  
NETMASK=255.255.0.0  
GATEWAY=172.16.0.1  
BROADCAST=172.16.255.255  
DHCP=no_  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
I /etc/sysconfig/network-scripts/ifcfg-eth0 [Modified] 9/9 100%
```

The terminal output shows the configuration of the Ethernet interface eth0 with a static IP address of 172.16.0.10 and various network parameters. The status at the bottom indicates the configuration was successfully applied.

Figure 5.2 — Assigning a Static IP to eth0

5.3 Restarting the Network Service

After restarting networking, the loopback and Ethernet interfaces come up successfully. The `ifconfig` output confirms `eth0` now holds `inet addr:172.16.0.10` with the correct broadcast and mask, and is UP, BROADCAST, RUNNING.



```
QEMU (Linux-1)

Network loopback interface lo is already down.
bringing up loopback interface lo ..... [ OK ]
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]
# ifconfig
eth0      Link encap:Ethernet  HWaddr 50:93:4C:01:AD:00
          inet addr:172.16.0.10  Bcast:172.16.255.255  Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

Figure 5.3 — Restarting the Network Service

5.4 Editing the Default Web Page

The server's default web page is edited to replace the placeholder content with a heading, a project title, and a footer link back to the project's GitHub repository — this is the page that will be served whenever the "Hotstar" server is reached over HTTP.

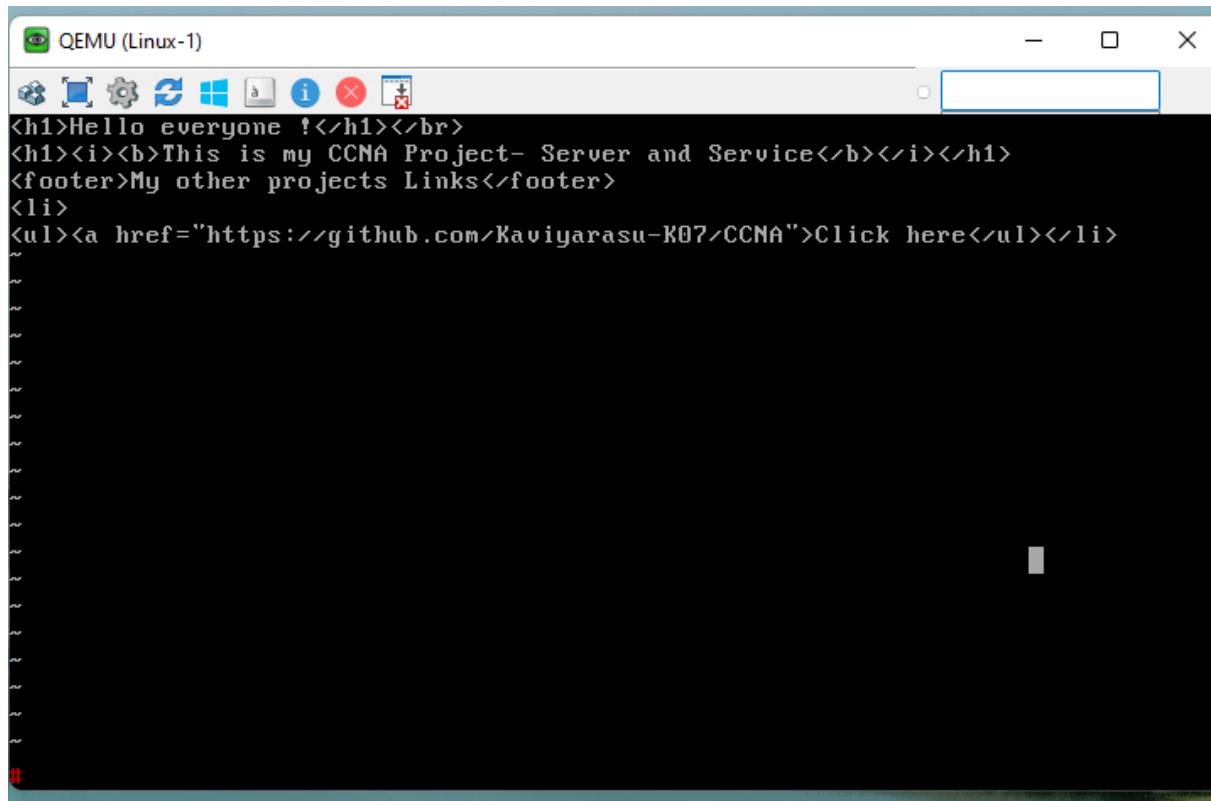


Figure 5.4 — Editing the Default Web Page

6. Netflix Server — Network Configuration

6.1 Assigning a Static IP on Winserver

In the IPv4 properties of the Windows Server's network adapter, the address is set to 172.16.0.20 with subnet mask 255.255.0.0 and default gateway 172.16.0.1, placing the server on the same datacenter subnet as Linux-1.

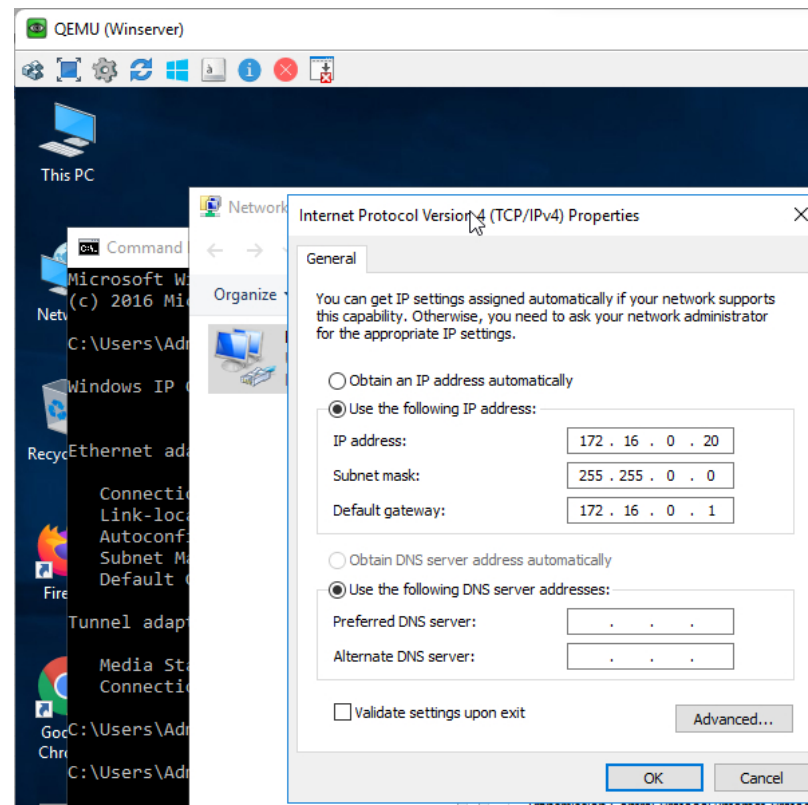
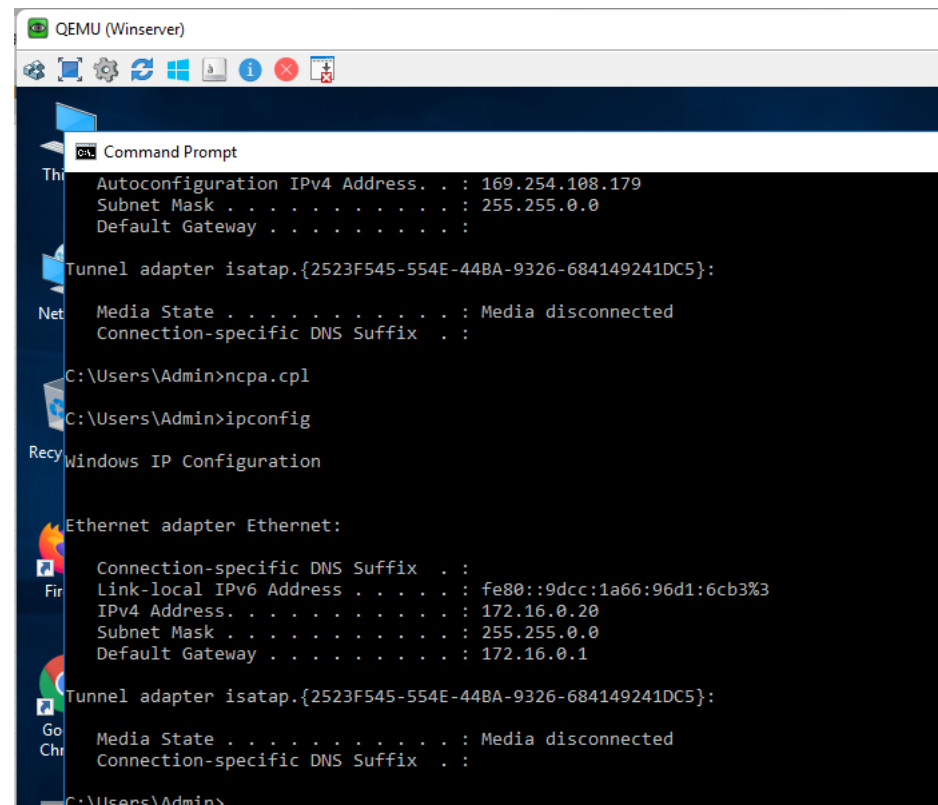


Figure 6.1 — Assigning a Static IP on Winserver

6.2 Verifying the Configuration with ipconfig

Running `ipconfig` from the command prompt confirms the Ethernet adapter now reports IPv4 address 172.16.0.20, subnet mask 255.255.0.0, and default gateway 172.16.0.1 exactly as configured.



```
QEMU (Winserver)
Command Prompt
The
Autoconfiguration IPv4 Address. . . : 169.254.108.179
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . :

Tunnel adapter isatap.{2523F545-554E-44BA-9326-684149241DC5}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

C:\Users\Admin>ncpa.cpl
C:\Users\Admin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::9dcc:1a66:96d1:6cb3%3
IPv4 Address. . . . . : 172.16.0.20
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 172.16.0.1

Tunnel adapter isatap.{2523F545-554E-44BA-9326-684149241DC5}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

C:\Users\Admin>
```

Figure 6.2 — Verifying the Configuration with ipconfig

6.3 Reviewing Network Connection Properties

The Network Connections control panel and the IPv4 properties dialog are reopened to double-check the saved settings before moving on to testing connectivity.

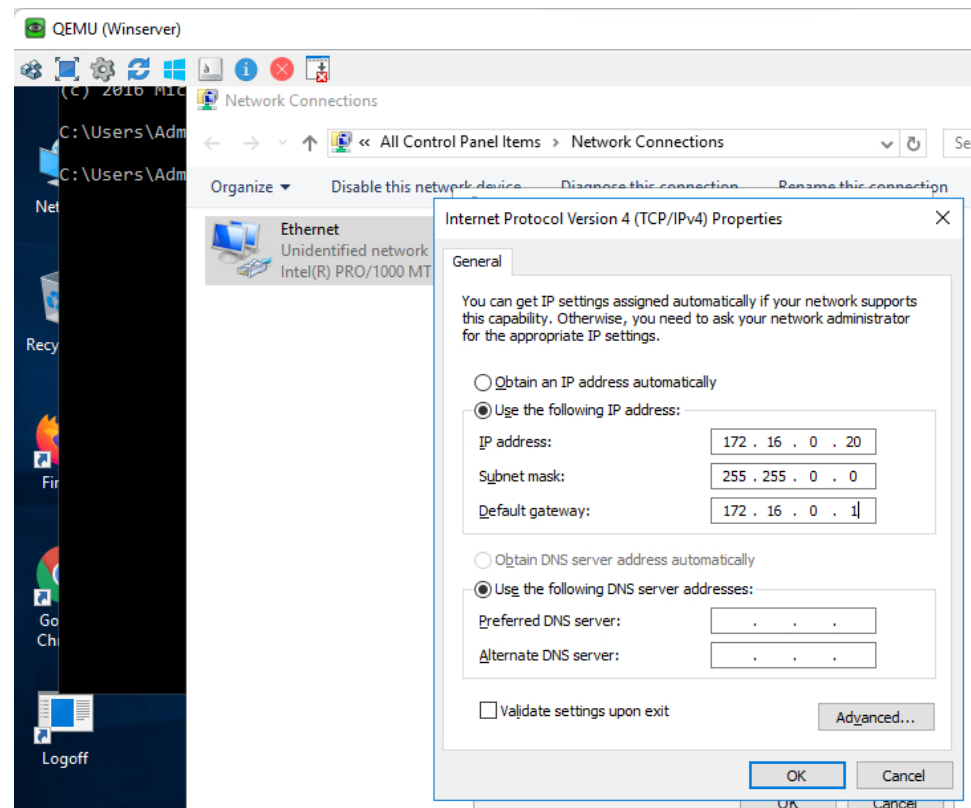


Figure 6.3 — Reviewing Network Connection Properties

7. Client PC Network Configuration

7.1 Assigning a Static IP to the Win Client

The client PC (representing an end user) is configured with IP address 50.0.0.10, subnet mask 255.0.0.0, and default gateway 50.0.0.1 — placing it on the client subnet behind the router's e0/2 interface.

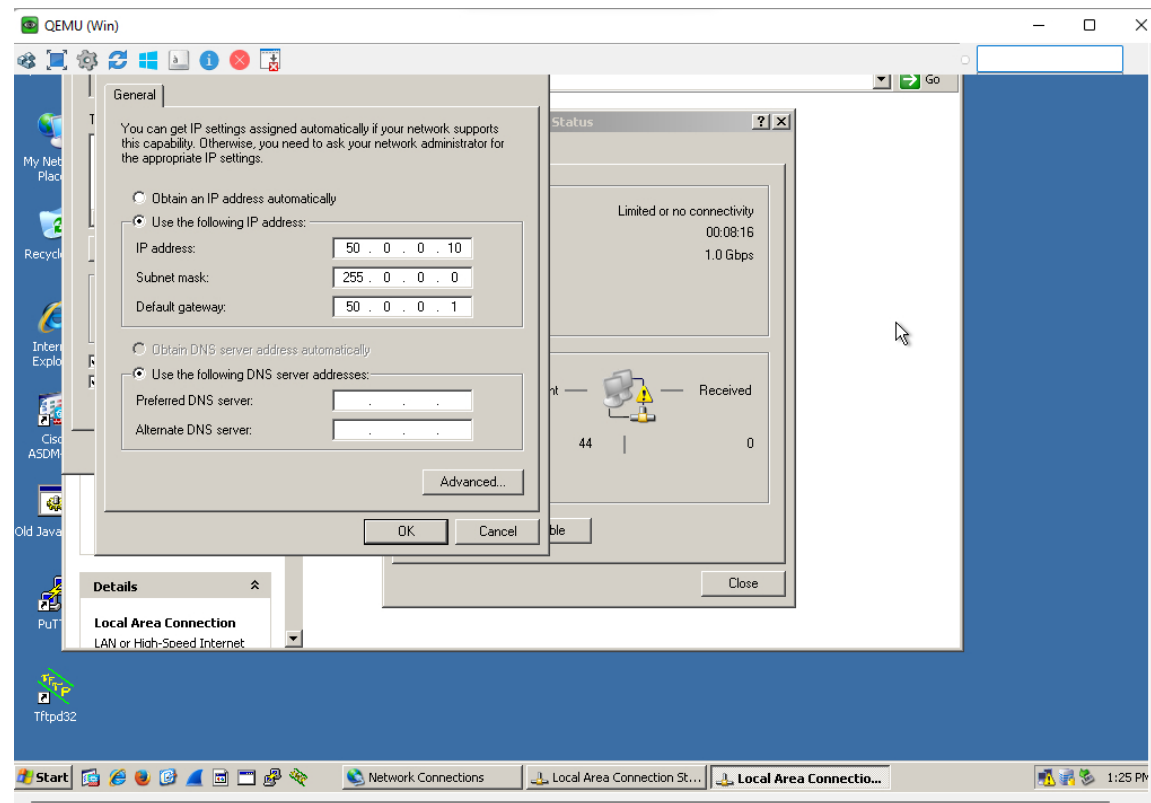
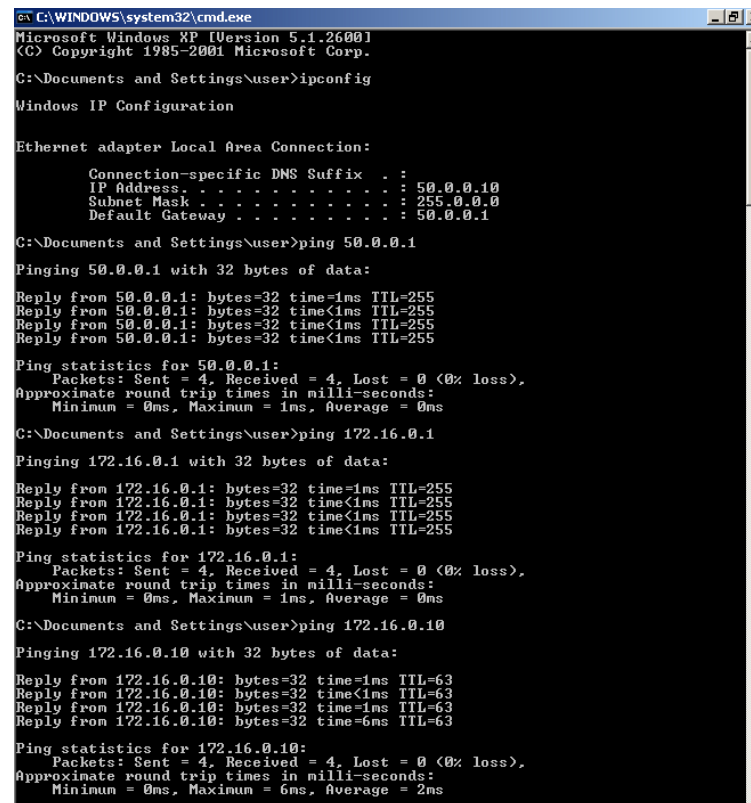


Figure 7.1 — Assigning a Static IP to the Win Client

8. Connectivity Testing — Ping Verification

8.1 Ping Test, Part 1

From the client, ping is used to verify reachability hop by hop: the default gateway 50.0.0.1 responds, the datacenter gateway 172.16.0.1 responds, and the Hotstar server at 172.16.0.10 responds with 0% packet loss — confirming the router is correctly forwarding traffic between the client and datacenter subnets.



```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 50.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 50.0.0.1

C:\Documents and Settings\user>ping 50.0.0.1

Pinging 50.0.0.1 with 32 bytes of data:

Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time<1ms TTL=255
Reply from 50.0.0.1: bytes=32 time<1ms TTL=255
Reply from 50.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 50.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>ping 172.16.0.1

Pinging 172.16.0.1 with 32 bytes of data:

Reply from 172.16.0.1: bytes=32 time=1ms TTL=255
Reply from 172.16.0.1: bytes=32 time<1ms TTL=255
Reply from 172.16.0.1: bytes=32 time<1ms TTL=255
Reply from 172.16.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

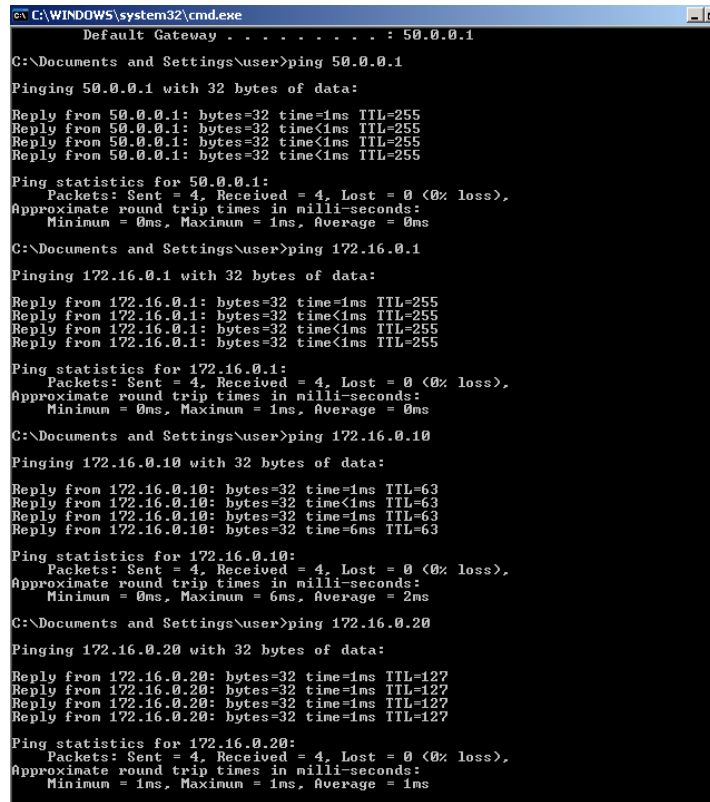
Reply from 172.16.0.10: bytes=32 time=1ms TTL=63
Reply from 172.16.0.10: bytes=32 time<1ms TTL=63
Reply from 172.16.0.10: bytes=32 time=1ms TTL=63
Reply from 172.16.0.10: bytes=32 time=6ms TTL=63

Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 2ms
```

Figure 8.1 — Ping Test, Part 1

8.2 Ping Test, Part 2

The same sequence is repeated and extended to 172.16.0.20, the Netflix (Windows Server) address. All four addresses along the path respond successfully, confirming full IP-level connectivity across all three subnets before any web traffic is tested.



```
C:\WINDOWS\system32\cmd.exe
Default Gateway . . . . . : 50.0.0.1
C:\Documents and Settings\user>ping 50.0.0.1
Pinging 50.0.0.1 with 32 bytes of data:
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time<1ms TTL=255
Reply from 50.0.0.1: bytes=32 time<1ms TTL=255
Reply from 50.0.0.1: bytes=32 time<1ms TTL=255
Ping statistics for 50.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Documents and Settings\user>ping 172.16.0.1
Pinging 172.16.0.1 with 32 bytes of data:
Reply from 172.16.0.1: bytes=32 time=1ms TTL=255
Reply from 172.16.0.1: bytes=32 time<1ms TTL=255
Reply from 172.16.0.1: bytes=32 time<1ms TTL=255
Reply from 172.16.0.1: bytes=32 time<1ms TTL=255
Ping statistics for 172.16.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Documents and Settings\user>ping 172.16.0.10
Pinging 172.16.0.10 with 32 bytes of data:
Reply from 172.16.0.10: bytes=32 time=1ms TTL=63
Reply from 172.16.0.10: bytes=32 time<1ms TTL=63
Reply from 172.16.0.10: bytes=32 time=1ms TTL=63
Reply from 172.16.0.10: bytes=32 time=6ms TTL=63
Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 2ms
C:\Documents and Settings\user>ping 172.16.0.20
Pinging 172.16.0.20 with 32 bytes of data:
Reply from 172.16.0.20: bytes=32 time=1ms TTL=127
Reply from 172.16.0.20: bytes=32 time=1ms TTL=127
Reply from 172.16.0.20: bytes=32 time=1ms TTL=127
Reply from 172.16.0.20: bytes=32 time=1ms TTL=127
Ping statistics for 172.16.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

Figure 8.2 — Ping Test, Part 2

9. Web Access Verification by IP Address

9.1 Accessing Hotstar via HTTP

Browsing to `http://172.16.0.10` from the client loads the custom page edited earlier on the Linux server — confirming the web service is running and reachable across the router purely by IP address.

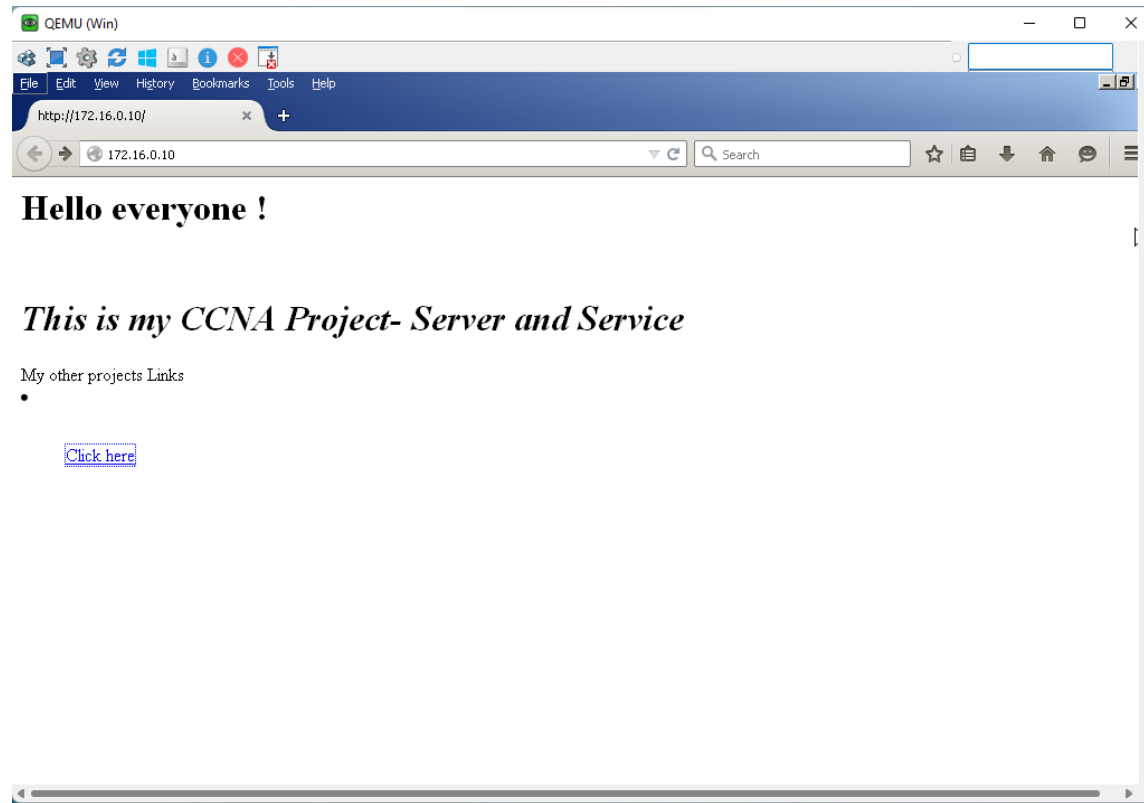


Figure 9.1 — Accessing Hotstar via HTTP

9.2 Accessing Netflix (IIS) via HTTP

In a second tab, browsing to `http://172.16.0.20` loads the default IIS welcome page from the Windows Server, confirming the IIS role installed earlier is active and serving content correctly.

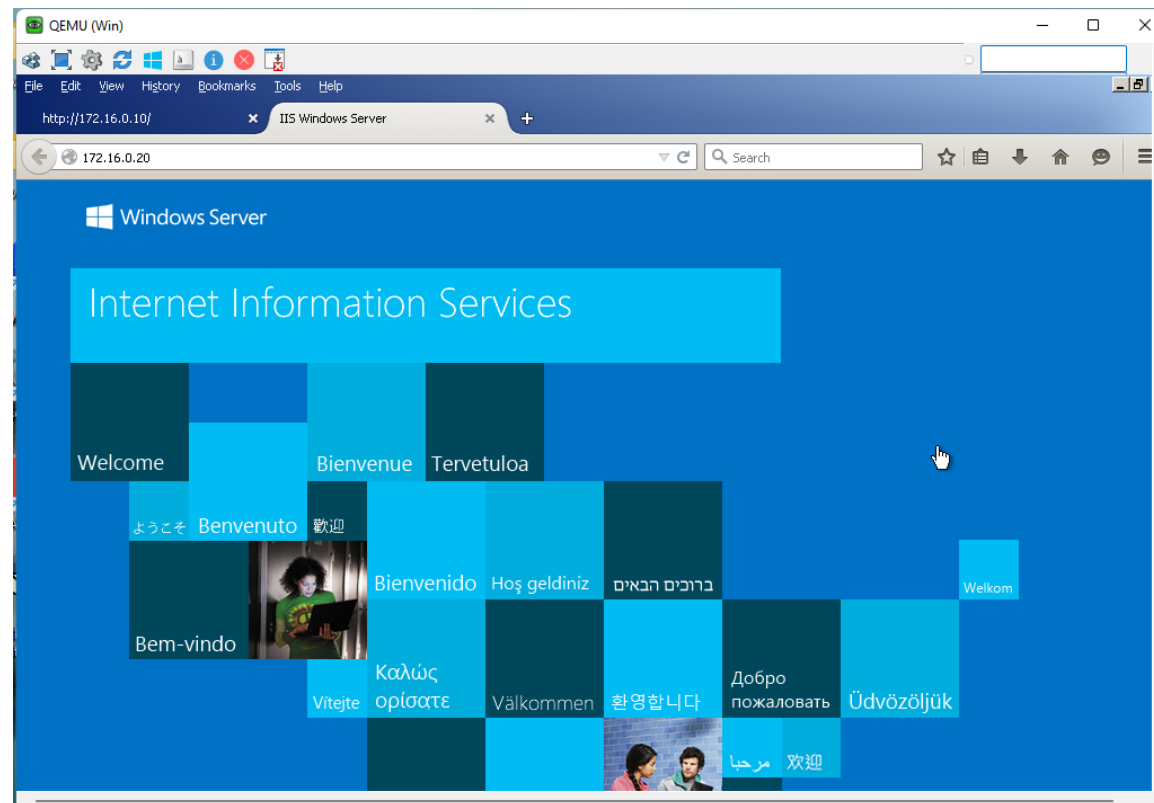


Figure 9.2 — Accessing Netflix (IIS) via HTTP

10. DNS Server Deployment

10.1 Installing the DNS Server Role

On the dedicated DNS virtual machine, the **DNS Server** role is selected in the Add Roles and Features Wizard. This role will host the forward-lookup zones needed to resolve the two custom domain names used in this project.

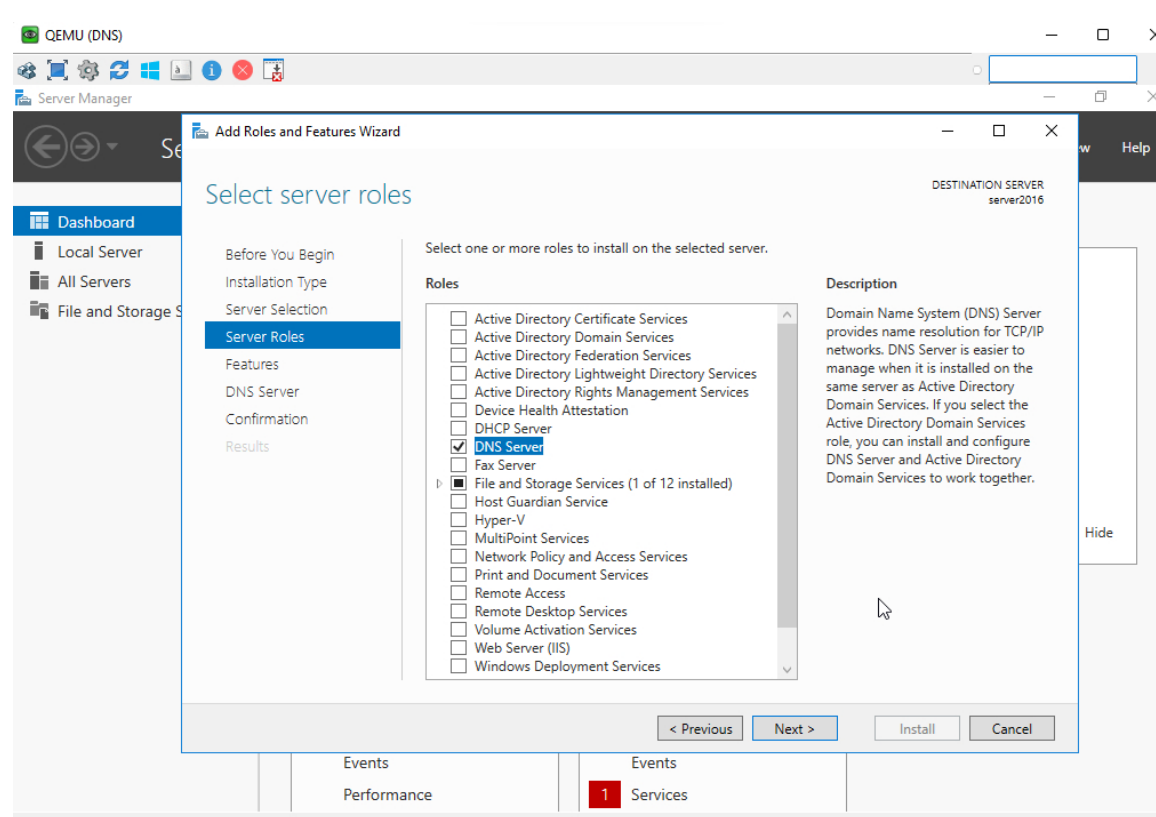


Figure 10.1 — Installing the DNS Server Role

10.2 DNS Role Installation Completed

The wizard confirms that the DNS Server role, along with the DNS Server Remote Server Administration Tools, installed successfully.

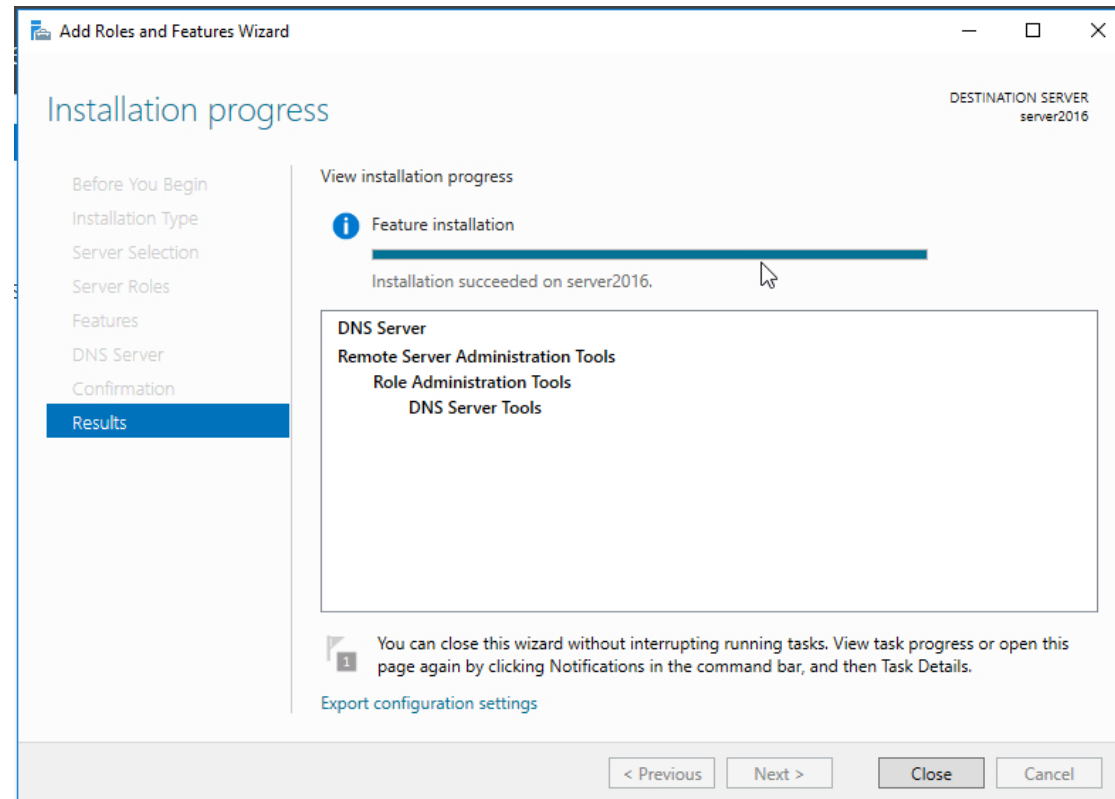


Figure 10.2 — DNS Role Installation Completed

10.3 Assigning a Static IP to the DNS Server

The DNS server's network adapter is configured with IP address 8.8.8.8, subnet mask 255.0.0.0, and default gateway 8.0.0.1, placing it on the DNS subnet behind the router's e0/1 interface.

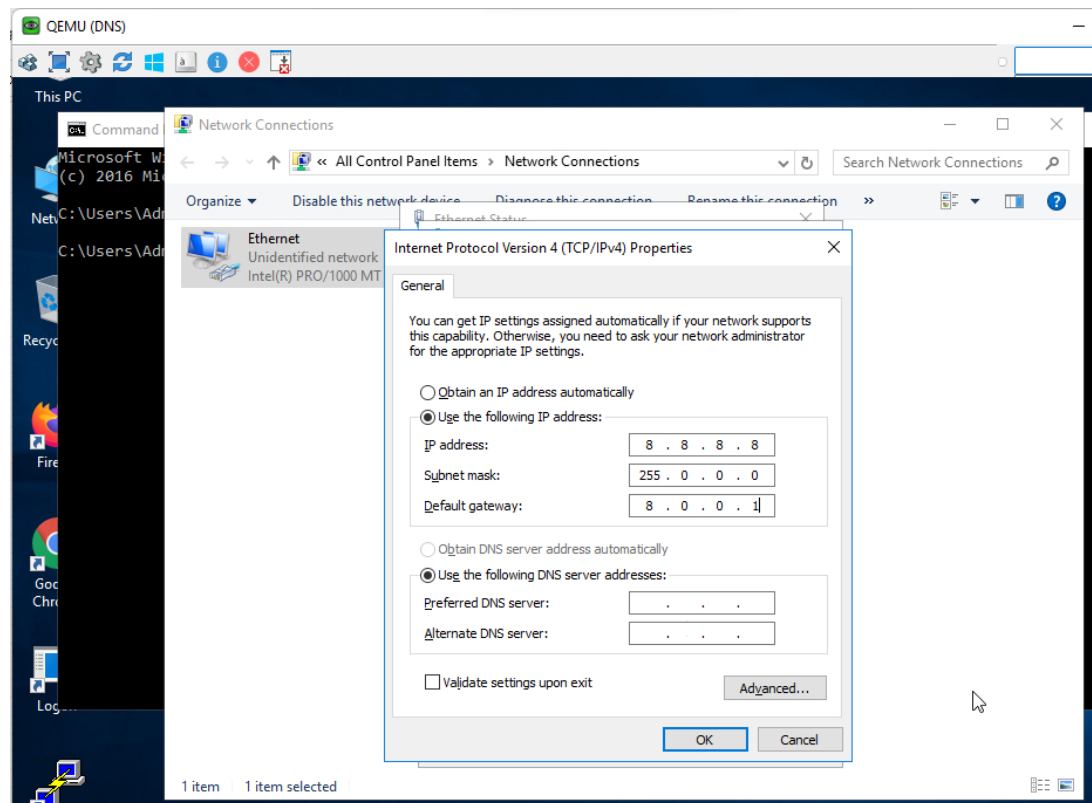


Figure 10.3 — Assigning a Static IP to the DNS Server

11. Client DNS Configuration

11.1 Configuring the Client's Preferred DNS Server

Back on the client PC, the IPv4 properties are updated so that the **Preferred DNS server** field points to 8.8.8.8 — the lab's own DNS server, not the public internet resolver — so all future name lookups are sent to it.

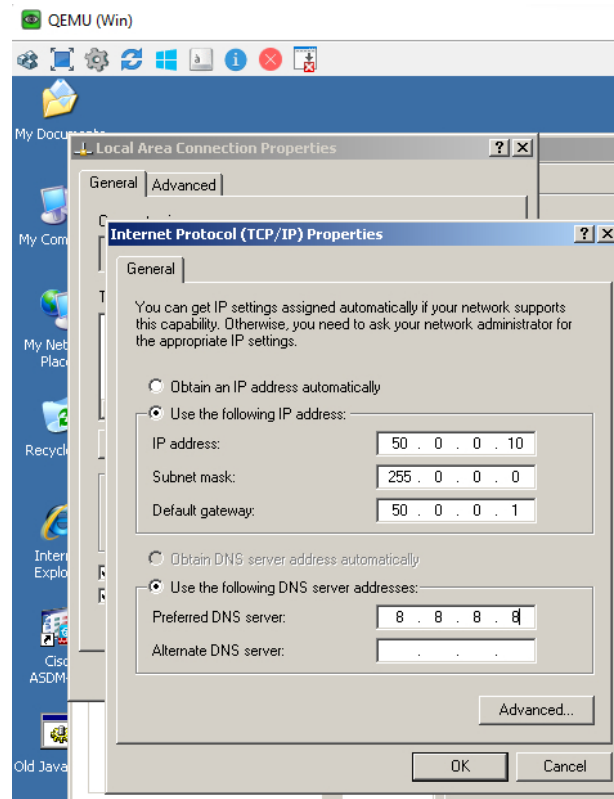


Figure 11.1 — Configuring the Client's Preferred DNS Server

12. Creating DNS Host Records

12.1 Creating the www.netflix.com Host Record

In DNS Manager, a forward lookup zone "netflix" is created under "com", and a New Host record named www is added with the fully qualified domain name www.netflix.com, pointing to IP address 172.16.0.20 — the Netflix (IIS) server.

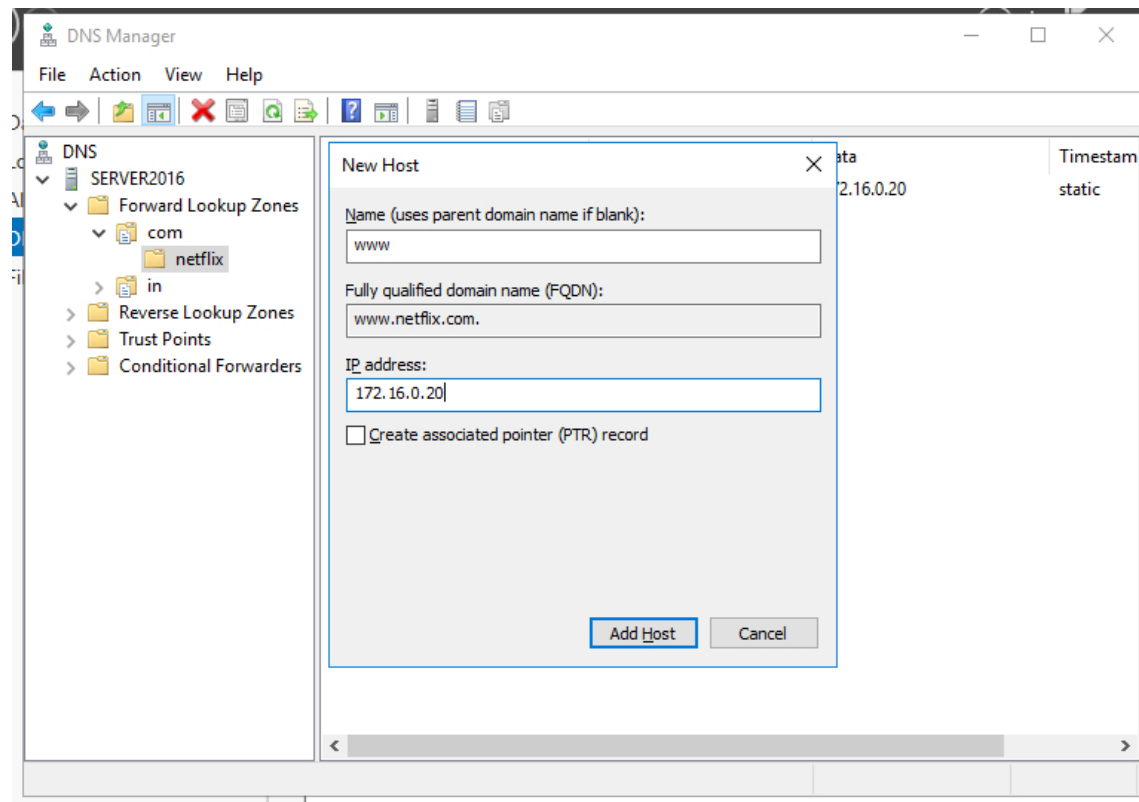


Figure 12.1 — Creating the www.netflix.com Host Record

12.2 Creating the www.hotstar.in Host Record

Similarly, a forward lookup zone "hotstar" is created under "in", with a host record for `www.hotstar.in` pointing to IP address 172.16.0.10 — the Hotstar (Linux) server.

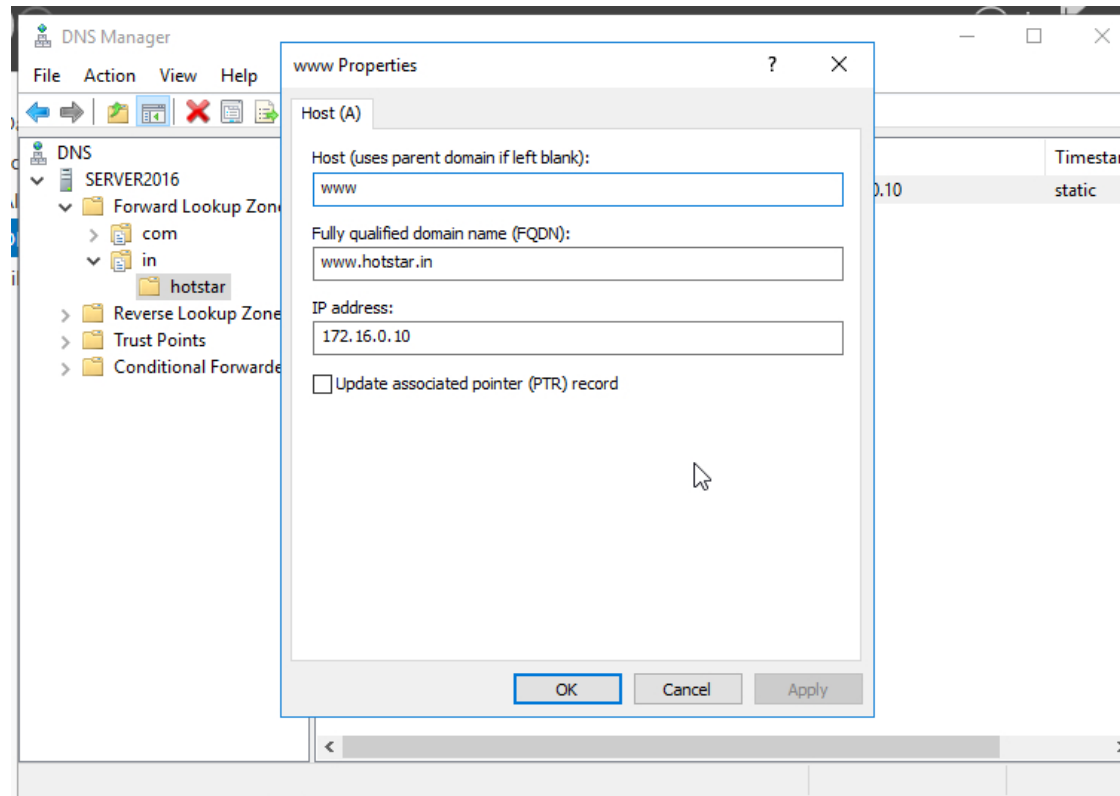


Figure 12.2 — Creating the www.hotstar.in Host Record

13. Final Verification by Domain Name

13.1 Resolving www.hotstar.in Successfully

With the client now using the lab DNS server, browsing to `www.hotstar.in` resolves correctly to 172.16.0.10 and loads the same custom page seen earlier — this time reached entirely by domain name instead of by IP address.

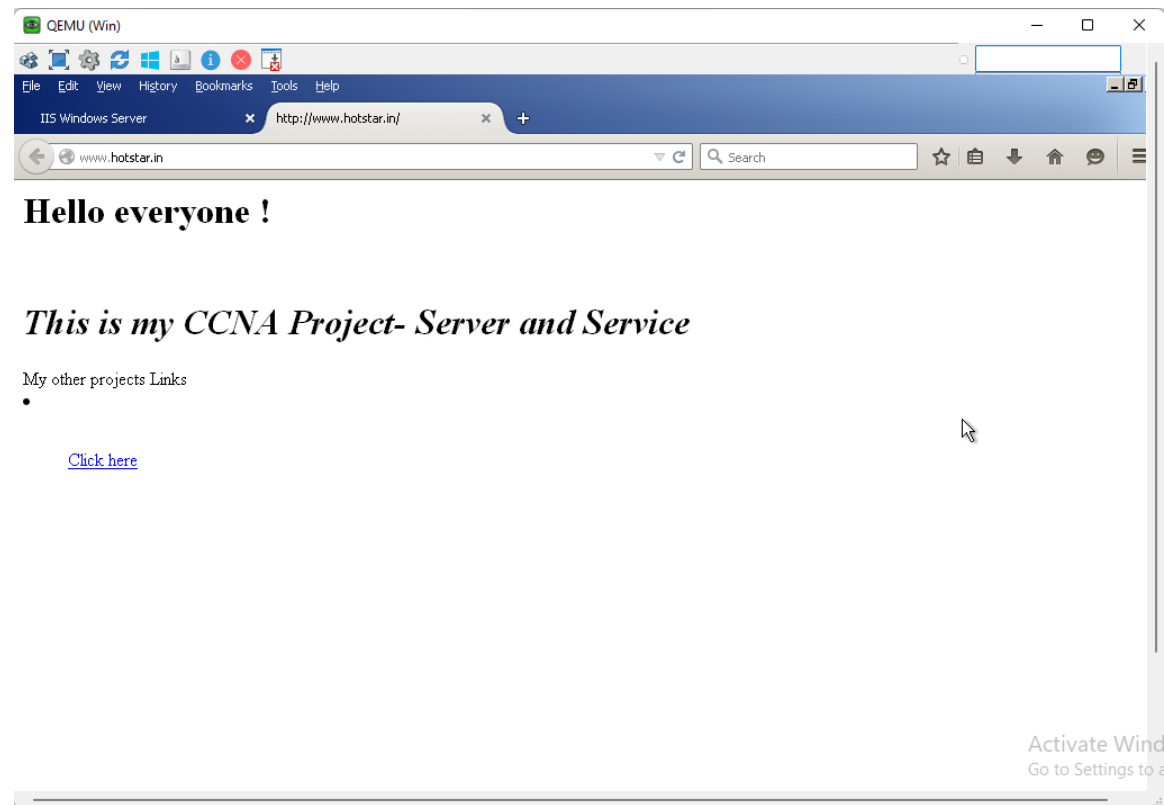


Figure 13.1 — Resolving www.hotstar.in Successfully

13.2 Testing www.netflix.com Resolution

The address bar is set to `www.netflix.com` to repeat the same test against the Netflix (IIS) host record created in the previous step, confirming the second DNS zone resolves to 172.16.0.20 as configured.

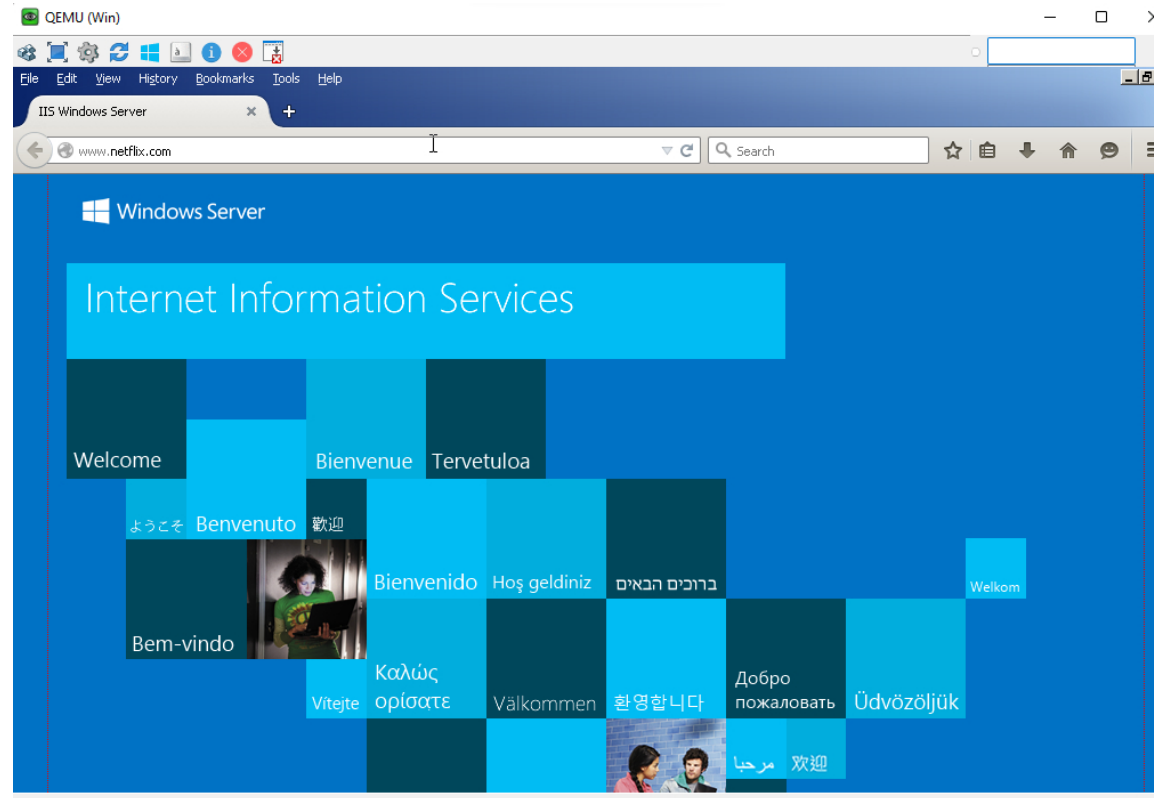


Figure 13.2 — Testing www.netflix.com Resolution

14. Conclusion

This lab brought together several core CCNA topics on a single router: static IP addressing across three subnets, routing between them, hosting web services on both a Windows Server (IIS) and a Linux host, and deploying a Windows DNS server with custom forward-lookup zones. Connectivity was verified in stages — first with ICMP ping, then with HTTP access by raw IP address, and finally with HTTP access by domain name once the client was pointed at the internal DNS server — demonstrating the full path from Layer 3 reachability up to application-layer name resolution.